

or a rotary dial to enter a telephone number for outgoing calls. Explore the small nitty-gritties. Why is a twisted pair preferred over an untwisted pair? How has the system been adapted for data communication such as telex, fax and Internet communication? Such probing will help you to acquire sufficient knowledge to overcome the interview with ease.

It is also beneficial to observe how this end-to-end analogue network has gradually evolved towards digital telephony. Starting with transmission networks modified with T1 carrier systems, use of later technologies such as SONET and fibre-optic transmission methods further facilitates digital transmission.

While today the end instrument remains analogue, the analogue signals are typically converted into digital after reaching the aggregation points: serving area interface or the central office. Digital loop carriers are often used, placing the digital network ever closer to the customer premises, relegating the analogue local loop to legacy status. Here, the network topologies, both infrastructure and *ad hoc*, demand attention. However, in real systems some customised versions are used depending on the cell concept and frequency of reuse. For example, the main aim of any cellular service provider—increasing of customer base within a limited frequency bandwidth—is also achievable by using a special type of infrastructure topology called multi-BS network configuration.

While dealing with the system design, you will face interferences. Two major kinds of interferences are co-channel interference and adjacent-channel interference. Try to get an idea of how to design a system with minimal interference. Expansion of the cellular network capacity is a governing factor behind system design. As a telecommunication engineer, you will handle it with the technologies related to change of cellular architecture, frequency allocation methodology, modem and access technology. Cell

splitting and sectoring are the other design techniques used for supporting the orderly growth of a cellular system. Basically, you have to use any specific technique depending on the channel assignment strategies: fixed channel allocation, dynamic channel allocation and hybrid channel allocation.

Access the mobility

Electronic communication began as a relatively simple concept. However, the increased demand for entertainment along with mobility is increasing its complexity. The voice-oriented first-generation (1G) wireless communication has evolved into the 3G system with personal communication service (PCS). Mahanagar Telephone Nigam Limited (MTNL) has finally launched its 3G mobile services, Jadoo, in limited areas of Delhi. This is the first 3G commercial launch by a telecom provider in India, though there was earlier a soft launch of 3G services in Delhi last year—incidentally by MTNL itself.

The increased complexity and advancements demand advance knowledge from the present-day technologists. Be aware of the key features of each generation of wireless communication systems. For example, the basic network in first generation follows analogue-based advanced mobile telephone system (AMPS). So you are expected to have an overview of channel controlling and call handling in AMPS.

Understand why did the successive generations come? Basic loopholes existing in the 1G system are poor battery performance, unavailability of channels, weak signal strength and poor handoff. So firms have realised the importance of digital technologies to improve battery performance as well as signal quality. 2G systems introduced narrow-band digital signals with global system for mobile communication (GSM). Here the digital cellular techniques like United States digital cellular (USDC) are adapted to provide better service.

Bear in mind, in supporting both the systems, cellular carriers are able to provide new customers with digital USDC phones while providing service to the existing AMPS analogue telephone customers. However, they are gradually replacing analogue base station with digital, channel by channel. This replacement is further facilitated due to high compatibility of USDC with AMPS. That's why USDC is often called digital AMPS (DAMPS).

Try to get some idea of the interim standards and related protocols. How significant are they? How do different digital control channels define the functionality of the USDC?

You should try to touch the most recent combination of mobility and entertainment—Internet access, music, TV and video, and other entertainment content. Know the path for 3G evolution to 3G/WCDMA. It is the air interface for one of the International Telecommunications Union's family of third-generation mobile communications systems. It is nice to know how it enables the continued support of voice, text and MMS services in addition to richer mobile multimedia services.

To sum up

Don't confine yourself to the textbooks. If possible, be a part of any live project which follows the recent trend. For example, in a new trend, global consumer electronics and mobile phone vendors are going green in India. Major players like Nokia, LG, Samsung and Haier, among others, are planning to introduce products that will be positioned on an environment-friendly platform, starting the trend of environment as a brand strategy in the Indian consumer electronics industry.

Bear in mind, with the rural India growth story unfolding, the telecom industry is likely to see tremendous growth in India's rural and semi-urban areas in the years to come. Are you ready to cope up with this diversity? ●

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